Exp. No. 12	Implementing AI for a real world problem (Fake news detection.)

Aim :

To implement an AI-based model that detects whether a given news article is real or fake using NLP and Machine Learning techniques.

Algorithm :

1. Data Collection :
   * Use a dataset containing labeled news articles (fake or real)
   * Example : fake_or_real_news.csv.
2. Data Preprocessing :
   * Clean text (remove punctuation, numbers, and stopwords).
3. Model Selection
   * Choose a classification algorithm
4. Training
   * Train the model on a training dataset.
5. Testing
   * Test the model on a separate test dataset.
   * Measure accuracy and performance metrics.
6. Prediction
   * Input a news headline/article → Predict whether it is Real or Fake.

Program :

```
import pandas as pd
```

```
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

df = pd.read_csv("fake_or_real_news.csv")

X = df['text']

y = df['label']

y=y.map({'FAKE':0, 'REAL':1})

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size
=0.2, random_state = 2)

vectorizer = TfidfVectorizer(stop_words = 'english', max_df = 0.7)
X_train_tfidf = vectorizer.fit_transform(X_train)
X_test_tfidf = vectorizer.transform(X_test)

model.fit(X_train_tfidf, y_train)

y_pred = model.predict(X_test_tfif)

print("✓ Accuracy:" accuracy_score(y_test, y_pred))

news = ["Government announces new policy for education
sector"]

new_tfidf = vectorizer.transform(news)
prediction = model.predict(news_tfidf)

if prediction[0] ==0:
    print("The news is : FAKE")
```

```
else:
    print ("The News is: REAL")
```

Sample Output:

Accuracy = 0.92

The News is : REAL

Result:

The Fake News Detection system was successfully implemented using Python, TF-IDF Vectorization, and Logistic Regression.